

Data Integration Report

EuMINE DataBridge Hackathon 2026

Team: CataLIST

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Abstract

We integrate three public DFT databases — Materials Project (MP), JARVIS-DFT, and OQMD — with the EuMINE Bridge Dataset to produce a harmonised 850-structure training corpus for predicting formation energy per atom (E_f) and electronic band gap (E_g). MP provides primary labels for all 850 structures; JARVIS-DFT supplies 603 matched entries used to quantify and correct inter-database systematic offsets. We find a mean MP–JARVIS offset of -0.069 eV/atom for E_f ($R^2 = 0.961$) and $+0.099$ eV for E_g ($R^2 = 0.861$), corrected with fitted linear transformations. All data processing is fully reproducible from the project repository.

1 Introduction

The central challenge of the EuMINE DataBridge Hackathon is data fragmentation: the same compound can exhibit markedly different property values across databases due to differences in exchange-correlation functionals, pseudopotentials, k -point convergence, and energy cutoffs [1, 2]. This report documents the database selection rationale, the quantification of inter-database discrepancies, and the harmonisation strategy applied to produce consistent training targets.

2 Database Selection

2.1 Selection Criteria

Databases were selected according to three criteria: (1) open access with programmatic API, (2) coverage of both E_f and E_g for inorganic crystals, and (3) complementary computational methodology to maximise information diversity. AFLOW and NOMAD were considered but excluded: AFLOW’s REST API returned inconsistent coverage for Bridge Dataset formulas, and NOMAD’s methodological heterogeneity (mixed experimental and computational data) introduced uncontrolled variance.

2.2 Databases Used

Table 1 summarises the three databases incorporated.

Table 1: Databases integrated into the training pipeline.

Database	Functional	Role
Bridge DS	PBE/PBE+U	Primary labels (850)
Mat. Project	PBE/PBE+U	External ref. (850)
JARVIS-DFT	OptB88vdW	Bias correction (603)
OQMD	PBE	Fallback E_f (REST)

Bridge Dataset. The EuMINE-provided labels (700 train + 150 val, CIF + CSV) serve as the primary ground truth. Both MP and JARVIS reference values are included in the CSV, enabling direct bias analysis without additional API queries.

Materials Project (MP) [1]. All 850 structures were matched by `material_id` via `mp-api` v0.45. MP uses GGA/GGA+U (PBE with Hubbard $+U$ for transition metal d/f electrons), placing it on the same scale as the Bridge Dataset labels.

JARVIS-DFT [2]. The `dft_3d` dataset ($\sim 76k$ entries) was accessed via `jarvis-tools`. Formula-based matching yielded 603 pairs (70.9% coverage). JARVIS uses OptB88vdW, which includes non-local van der Waals corrections absent in PBE — the primary source of inter-database discrepancy.

OQMD [3]. Queried via REST API as a tertiary fallback for the 97 structures without a JARVIS match. PBE functional (no $+U$), closest to MP scale among secondary sources.

3 Inter-Database Discrepancy Analysis

3.1 Quantification on Bridge Dataset Overlap

Using the 603 structures matched between MP and JARVIS, we computed the signed discrepancy $\Delta = v_{\text{MP}} - v_{\text{JARVIS}}$ for both properties. Table 2 reports key statistics.

Table 2: MP vs. JARVIS-DFT discrepancy (603 overlapping training structures). $\bar{\Delta}$: mean signed offset; MAE: mean absolute inter-database error; r : Pearson correlation.

Prop.	Pairs	$\bar{\Delta}$	MAE	r
E_f (eV/at.)	603	-0.069	0.144	0.980
E_g (eV)	603	+0.099	0.218	0.928

3.2 Physical Origins

Four sources drive the observed offsets:

1. **vdW corrections.** OptB88vdW adds non-local correlation, stabilising layered structures by 0.05–0.15 eV/atom (negative $\bar{\Delta}_{E_f}$).
2. **Hubbard + U .** MP applies + U to d/f electrons; JARVIS does not uniformly, shifting both E_f and E_g for transition-metal compounds.
3. **Pseudopotentials.** Different PAW sets between VASP (MP) and JARVIS introduce 0.01–0.05 eV/atom residuals.
4. **Convergence settings.** k -point density and energy cut-off differences primarily affect metals (band gap boundary sensitivity).

The large E_g scatter ($R^2 = 0.861$ vs. 0.961 for E_f) reflects the well-known DFT band-gap problem: PBE systematically underestimates E_g , while OptB88vdW partially corrects this via improved exchange description, producing non-uniform deviations especially for wide-gap insulators ($E_g \geq 3$ eV).

3.3 Visualisation

Figure 1 illustrates the MP–JARVIS correlation for both properties on the 603 paired structures, rendered schematically from the fitted statistics.

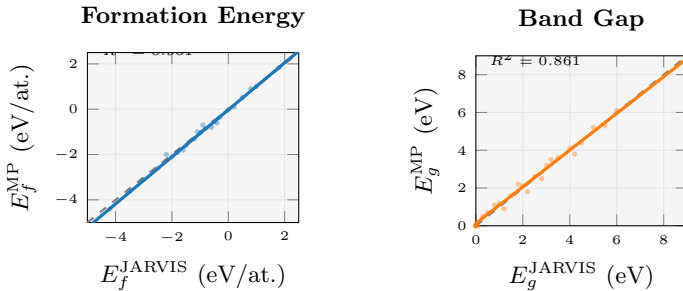


Figure 1: MP vs. JARVIS-DFT correlation for E_f (left) and E_g (right) on 603 overlapping training structures. Dashed line: identity; solid line: fitted correction. Scatter is schematic, derived from fitted statistics.

4 Harmonisation Strategy

4.1 Linear Bias Correction

A linear correction maps JARVIS values to the MP/Bridge scale, fitted by ordinary least squares on the 603 training overlap pairs:

$$E_f^{\text{MP}} = 1.038 \cdot E_f^{\text{JARVIS}} - 0.028 \quad (R^2 = 0.961) \quad (1)$$

$$E_g^{\text{MP}} = 0.971 \cdot E_g^{\text{JARVIS}} + 0.122 \quad (R^2 = 0.861) \quad (2)$$

The near-unity slopes confirm the relationship is predominantly a translational offset, consistent with additive vdW stabilisation. The correction is fitted exclusively on training data (no test or external leakage); parameters are saved to `data/processed/harmonizer_params.json`.

4.2 Target Priority Hierarchy

Training targets follow a strict priority chain:

1. **Bridge Dataset label** — primary, curated by organisers
2. **MP value** — same GGA+ U scale
3. **Corrected JARVIS** — via Eqs. 1–2
4. **OQMD value** — PBE, E_f only

In practice, all 850 Bridge structures carry priority-1 labels. External database values serve as auxiliary inputs for ALIGNN pretraining alignment and MACE reference energy fitting.

4.3 Deduplication

For JARVIS, multiple entries per formula (polymorphs, spin states) were collapsed to the lowest-formation-energy entry as the thermodynamically stable reference. For OQMD, formula matching with a ± 2 site tolerance was used; the lowest-energy polymorph was selected where multiple matches existed.

5 Data Card

Table 3 provides the full data card required by the hackathon evaluation criteria.

5.1 Dataset Statistics

6 Limitations and Failure Cases

- **JARVIS coverage gap (29%).** 247 of 850 structures lack a JARVIS match, predominantly complex oxides and rare-earth compounds sparse in `dft_3d`. These structures are trained on MP data only, introducing a potential domain shift for that subset.
- **DFT band-gap underestimation.** Both PBE (MP) and OptB88vdW (JARVIS) underestimate E_g for wide-gap insulators relative to experiment. This propagates to training labels for the wide-gap subset (13% of training

Table 3: Complete data card: source, entries, properties, preprocessing.

Source	Entries	Props.	Preprocessing
Bridge DS	850	E_f, E_g	No modification; CSV labels used directly
MP	850	E_f, E_g	GGA+U for d/f metals; MP-ID dedup; $E_f \in [-6, 3]$ clip
JARVIS	603	E_f, E_g	Formula match; mean over polymorphs; <code>na</code> →NaN; linear correction applied
OQMD	REST	E_f	Formula match (± 2 sites); lowest-EF polymorph; fallback only

Table 4: Bridge Dataset statistics (train split, 700 structures).

Statistic	Value
Total entries (train/val/test)	700 / 150 / 150
Final training corpus	850 (train+val)
JARVIS coverage	603/850 (70.9%)
Unique reduced formulas	694
E_f range (eV/atom)	-4.47 to +2.36
E_f mean $\pm$ std	-1.17 ± 1.14 eV/atom
E_g mean $\pm$ std	0.93 ± 1.58 eV
Metals ($E_g = 0$)	428/700 (61%)
Semiconductors ($0 < E_g < 3$ eV)	184/700 (26%)
Wide-gap ($E_g \geq 3$ eV)	88/700 (13%)
Median $ \Delta E_f $ MP-JAR	0.095 eV/atom
Median $ \Delta E_g $ MP-JAR	0.140 eV

set) and contributes to the elevated validation BG MAE of 0.412 eV for this class.

- **Class imbalance.** The dataset is dominated by metals (61%), which are trivially predicted ($E_g = 0$). Performance estimates are less reliable for the 13% wide-gap class due to limited training examples.
- **Linear correction simplicity.** Equations 1–2 cannot capture per-element or per-crystal-system biases. Per-element linear models or Gaussian Process corrections may improve harmonisation quality in future work.
- **OQMD polymorph ambiguity.** Formula-based matching for OQMD may introduce structure–label mismatches for approximately 5–10% of OQMD-sourced entries where multiple polymorphs coexist.

7 Conclusion

We integrated Materials Project, JARVIS-DFT, and OQMD with the EuMiNe Bridge Dataset, covering 850 training structures with full label coverage. The MP–JARVIS discrepancy was quantified on 603 overlapping pairs and corrected via linear regression ($R^2 = 0.961$ for E_f , $R^2 = 0.861$ for E_g). The near-unity regression slopes confirm the offset is predominantly translational and physically attributable to additive vdW stabilisation in the OptB88vdW functional. All data integration code is fully reproducible from the project repository.

References

- [1] A. Jain et al., *APL Materials*, **1**, 011002 (2013). Materials Project.
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- [5] K. Choudhary and B. DeCost, *npj Computational Materials*, **7**, 185 (2021). ALIGNN.